Final
May 17th

For each question, please provide details or middle steps in addition to final answers.

1. (15 points) Consider a DTMC $X = \{X_n : n = 0, 1, 2, \ldots\}$ on state space $S = \{a, b\}$. The transition probability matrix is

$$P = \begin{pmatrix} .2 & .8 \\ .5 & .5 \end{pmatrix}.$$

Each time the DTMC visits state a, the system loses \$10. Each time the DTMC visits state b, the system wins \$20. Denote

$$g(a) = -\$10 \quad \text{and} \quad g(b) = \$20.$$

(a) (5 points) Compute

$$\mathbb{P}\{X_2 = b, X_1 = a | X_0 = a\} \text{ and } \mathbb{P}\{X_2 = b | X_0 = a\}.$$

(b) (5 points) Let $\beta = 1.0$. Compute the expected total discounted profit in the first three timeslots, including timeslot 0. Namely,

$$\mathbb{E}\Big[\sum_{n=0}^{2} \beta^n g(X_n) | X_0 = a\Big].$$

(c) (5 points) Let $\beta = 0.5$. Compute

$$\mathbb{E}\Big[\sum_{n=0}^{\infty} \beta^n g(X_n) | X_0 = a\Big].$$

(You may leave the final answer in a simplified matrix expression. No need to compute the inverse of matrix.)

2. (15 points) A store sells a perishable product B with two week lifetime. Suppose that the weekly demand D for product B is i.i.d with the distribution

d	5	10	15	20
$\mathbb{P}\{D = d\}$	1/2	1/6	1/6	1/6

At the end of the sales day on Friday, unsold items are sorted. If the "age" of an item is one (sales) week, i.e. it arrived on Monday morning 4 days ago, then it is used in the following week selling at a price of \$150. Otherwise, it is discarded without any salvage value.

Consider the following ordering policy. Each Friday evening, if the number of one-week old items is strictly less than 10, order enough to bring the following Monday morning's (total) inventory to 20. Otherwise, do not order. Each item costs \$100 to order, with no fixed ordering cost.

Each week, customer demand is first fulfilled by new items. When the new items run out, additional demand is fulfilled by old items. Unsatisfied demand is lost. New items sell at \$200 per item. There is no holding cost.

(a) (5 points) Let X_n be the number of items on Monday morning of week n, regardless of the age of items. Is $X = \{X_n : n = 0, 1, ...\}$ is a DTMC? If so, specify its state space and the transition matrix. Show your work to support your claims.

(b) (5 points) Let Y_n be the number of one-week old items at the end of week n Friday. Is $Y = \{Y_n : n = 0, 1, \ldots\}$ a DTMC? If so please specify its state space and the transition matrix.

(c) (5 points) Assume you are given the stationary distribution of some appropriate Markov chain. In other words, if you said that $\{X_n\}$ or $\{Y_n\}$ (or both) is a Markov chain, its stationary distribution is assumed to be given to you. Find the long-run average profit week per week in terms of the stationary distribution. (You do not need to actually compute the stationary distribution, just use the dummy variables π_i, where i ranges across all possible values in the state space of your Markov chain. You do need to compute the numerical values of other quantities.)

3. (20 points) Assume the DTMC has state space $S = \{1, 2, 3, 4, 5, 6, 7\}$ and transition probability matrix

$$P = \begin{pmatrix} 0.2 & 0.8 & 0 & 0 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0.8 & 0 & 0.2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

(a) (2 points) Draw the transition diagram.

(b) (2 points) Is the DTMC irreducible? If not, specify *all* communicating classes.

(c) (2 points) For each state $i \in S$, find its period.

(d) (4 points) For each state $i \in S$, find $\mathbb{E}(T_{i,i})$, the expected time for the DTMC to return state i, starting from i. Adopt the convention that $\mathbb{E}(T_{i,i}) = \infty$ if state i is transient.

(e) (4 points) Find *two* stationary distributions of the DTMC.

(f) (6 points) Find the 7×7 matrix

$$\lim_{n\to\infty} P^n.$$

When an entry $\lim_{n\to\infty} P^n_{ij}$ does not exist, use "n" to mark that entry and explain why the entry does not exist.

4. (20 points) A call center has two agents, John and Mary. It has three phone lines. An incoming call that finds all lines are occupied receives a busy signal and is rejected. An accepted call finds all agents busy waits in the queue. While waiting (listening to music), the caller may abandon without service. When a caller's waiting time exceeds her patience time, the caller abandons without service. The patience times are i.i.d exponentially distributed with mean 2 minutes. John's processing times of calls are i.i.d exponentially distributed with mean 2 minutes. Mary's processing times of calls are iid exponentially distributed with mean 1 minute. The arrival process is Poisson with rate $\lambda = 1$ call per minute. An arrival to an empty system is always processed by John.

(a) (10 points) Formulate a CTMC to describe the system by specifying *all three* inputs (holding rates + jump matrix, generator matrix, transition rate diagram) for this CTMC.

(b) (5 points) Find the long-run proportion of customers that get rejected by the system and explain why.

(c) (5 points) Find the long-run average time in the system for each accepted call.

5. (12 points) Consider a special two-station open Jackson network. Each station has a single server and an infinite capacity waiting area. External arrivals to station 1 follow a Poisson process with rate α, and service times at station i are iid, following exponential distribution with rate μ_i, $i = 1, 2$. After being served at station 1, each customer goes to station 2, and after being served at station 2, the customer goes back to station 1 with probability 60% and leaves the system with probability 40%.

Assume that

$$(2.5)\alpha < \mu_1 \text{ and } (2.5)\alpha < \mu_2.$$

Let $X_i(t)$ be the number of customers at station i at time t, $i = 1, 2$. Then $X = \{(X_1(t), X_2(t)), t \geq 0\}$ is a continuous time Markov chain.

Prove that X is positive recurrent by proving that (a) (2 points) X is irreducible, and (b) (10 points) X has a stationary distribution. For (b), do the following three steps: (i) write the flow balance equation for *each state*, (ii) guess a form of the stationary distribution, and (iii) verify that the guessed stationary distribution does satisfy the flow balance equation for each state.

6. (18 points) Let $\xi_1, \xi_2, ...$ be i.i.d. with $\mathbb{P}(\xi_i = 1) = p$ and $\mathbb{P}(\xi_i = -1) = 1 - p$, where $1/2 < p < 1$. Consider the asymmetric simple random walk $\{S_n\}$ defined by $S_n = \xi_1 + \cdots + \xi_n, n \geq 1$. Define first hitting time $T_x = \inf\{n : S_n = x\}$ for any integer x. Let $a < 0 < b$ be two integers.

   (a) (4 points) Define $\phi(x) = \{(1-p)/p\}^x$. Prove that $\phi(S_n)$ is a martingale with respect to $\xi_1, \cdots, \xi_n$.

   (b) (4 points) Prove that $\mathbb{P}(T_a < T_b) = \frac{\phi(b)-\phi(0)}{\phi(b)-\phi(a)}$.

   (c) (4 points) Prove that $\mathbb{P}(T_a < \infty) = \{(1-p)/p\}^{-a}$ and $\mathbb{P}(T_b < \infty) = 1$.

   (d) (6 points) Prove that $\mathbb{E}T_b = b/(2p-1)$. (Hint: find some constant c such that $S_n - cn$ is a martingale.)